

Tratamiento Digital de Imágenes en JavaScript



Javier Velez Reyes

Madrid JS · Septiembre 2026

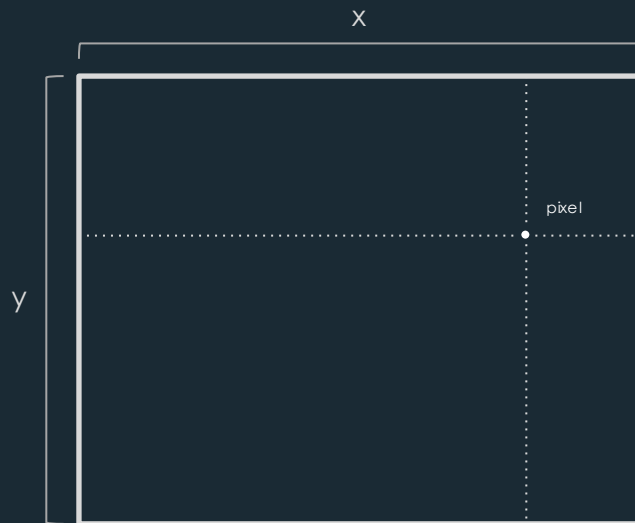
Introducción



tinyurl.com/254ztxro

Introducción

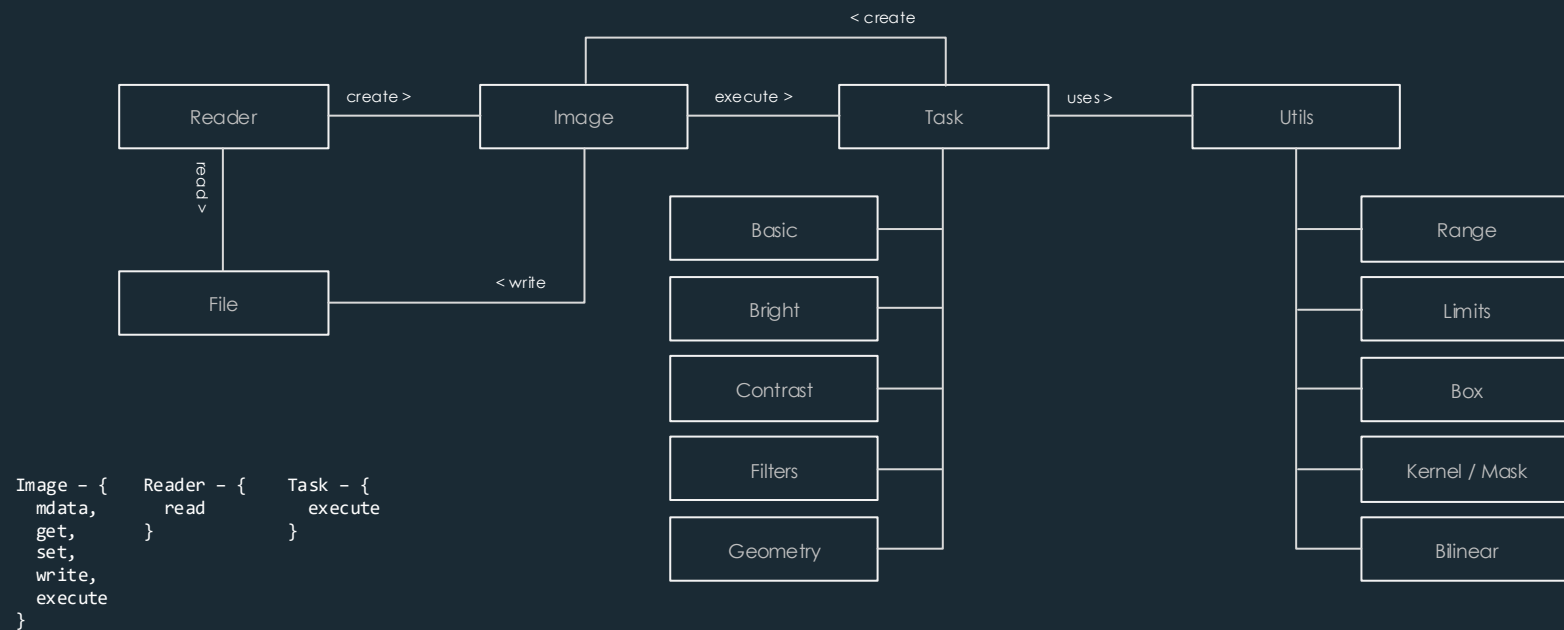
Imágenes & Información



```
Pixel (x, y) - {  
  red  : [0 .. 255],  
  green: [0 .. 255],  
  blue : [0 .. 255],  
  alfa : [0 .. 255]  
}
```

Introducción

Framework Imágenes en JS



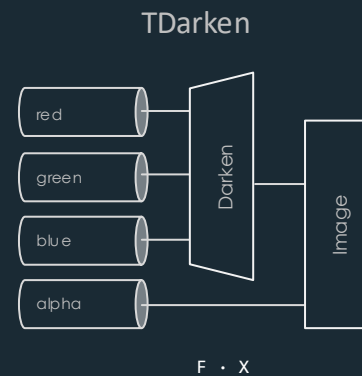
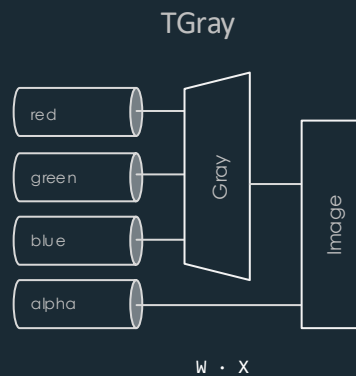
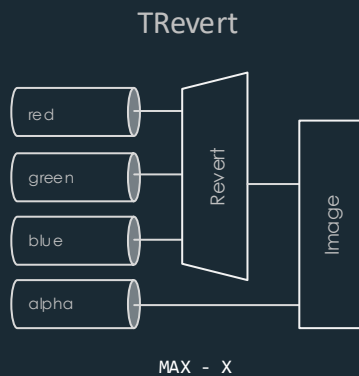
Introducción

Catálogo de Operaciones

		Propósito					
		Corrección	Mejora	Codificación	Efectos	Transformar	Análisis
Alcance	Punto	X	X	X	X		X
	Vecindad	X	X	X	X		X
	Frecuencia	X	X	X	X		X
	Geometría	X			X	X	

Tratamiento de Imágenes

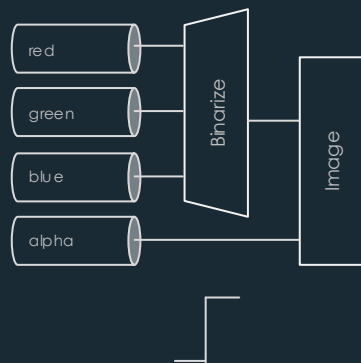
Operaciones Básicas



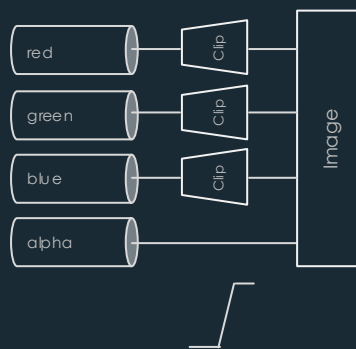
Tratamiento de Imágenes

Operaciones de Contraste

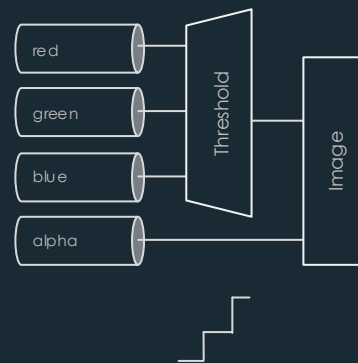
TBinarize



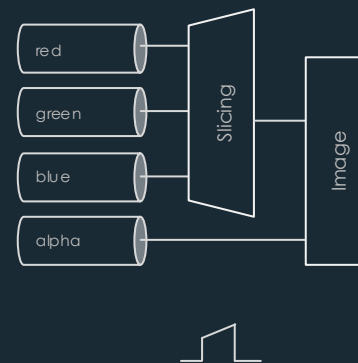
TClipping



TThreshold

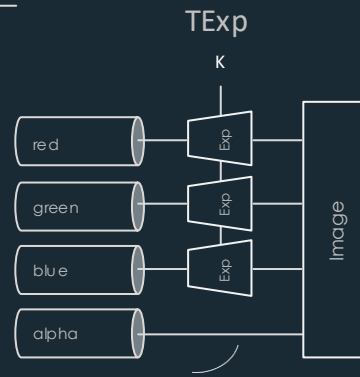
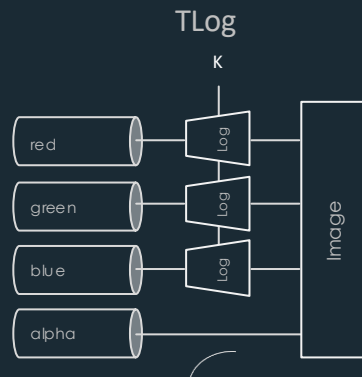
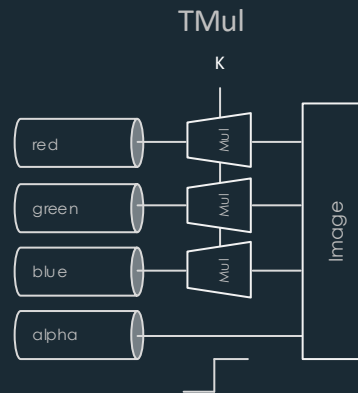
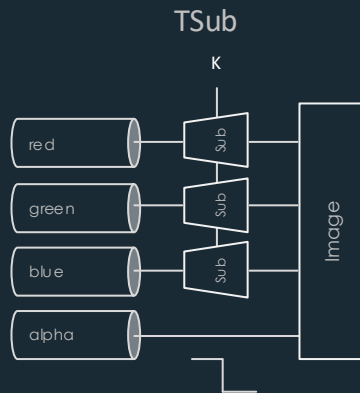
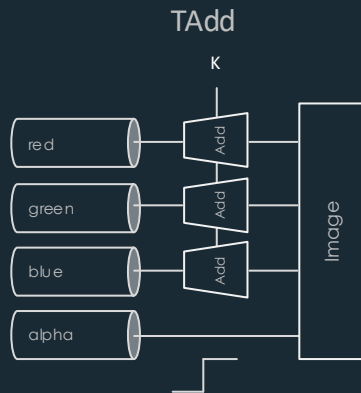


TSlicing



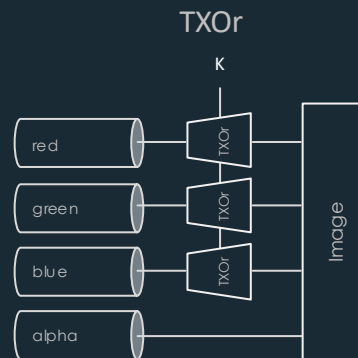
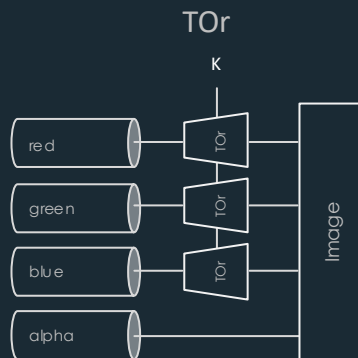
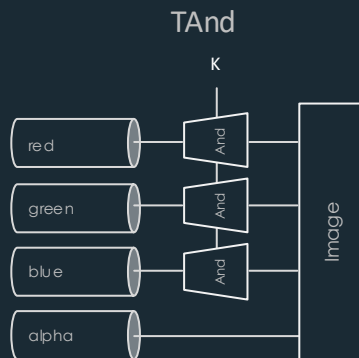
Tratamiento de Imágenes

Operaciones de Brillo · Aritméticas



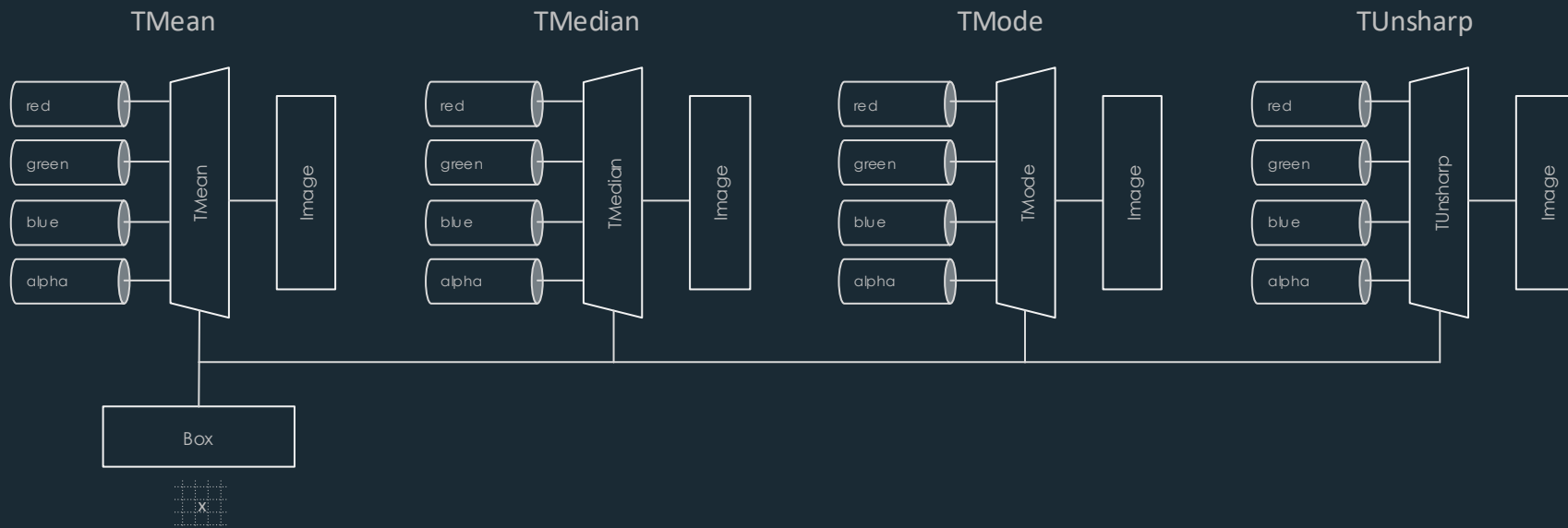
Tratamiento de Imágenes

Operaciones de Brillo · Lógicas



Tratamiento de Imágenes

Operaciones de Filtro · Estadísticos

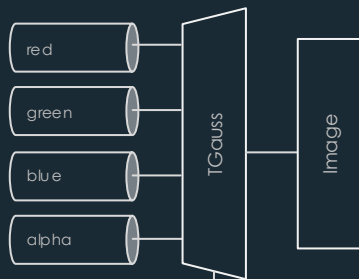


Tratamiento de Imágenes

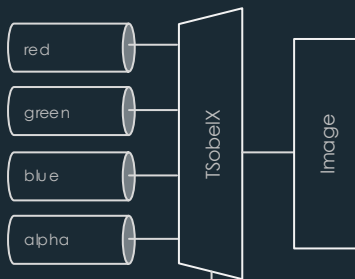
Operaciones de Filtro · Kernel

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

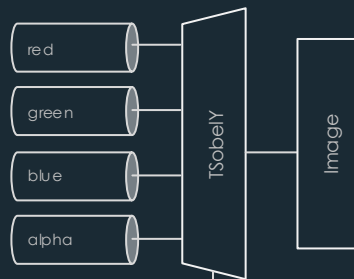
TGauss


$$\begin{bmatrix} -1 & 1 & 1 \\ -2 & 1 & 1 \\ -1 & 1 & 1 \end{bmatrix}$$

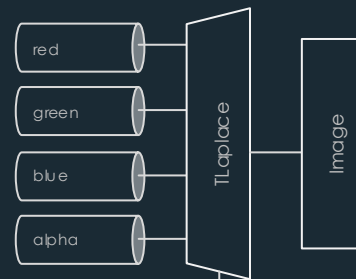
TSobelX


$$\begin{bmatrix} -1 & -2 & -1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

TSobelY

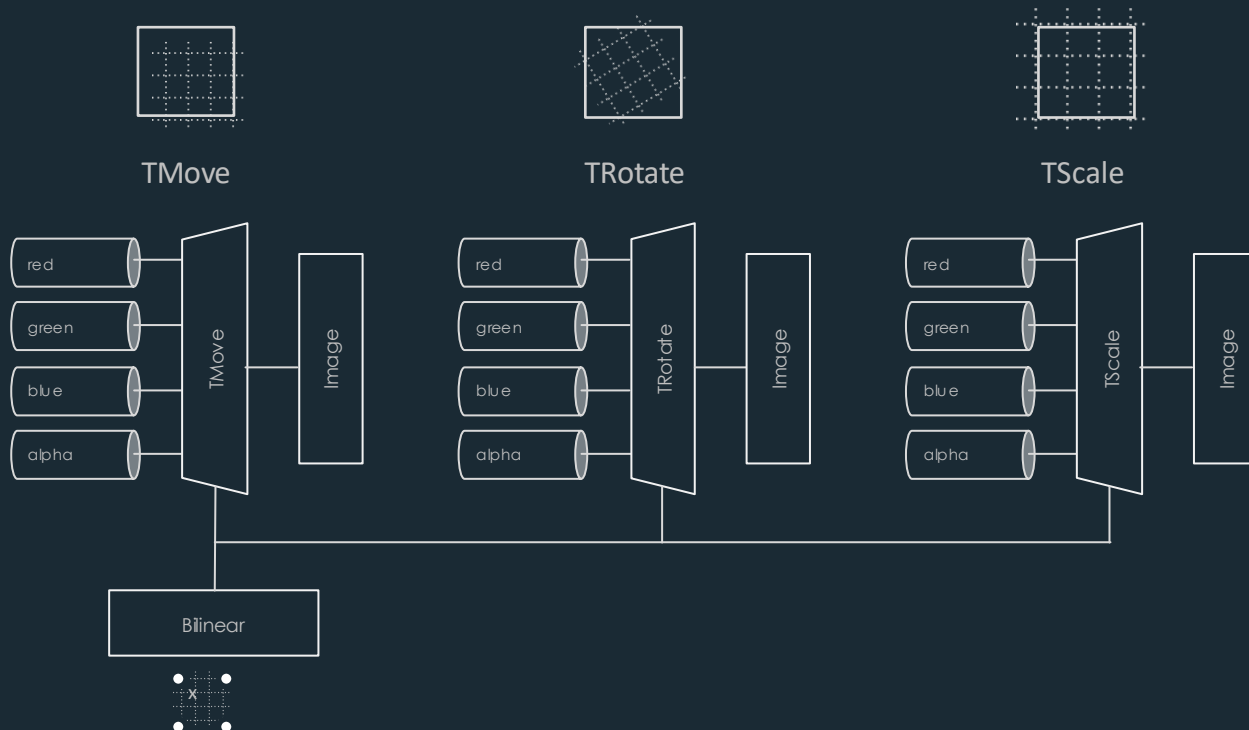

$$\begin{bmatrix} -1 & -1 & -1 \\ -1 & 8 & -1 \\ -1 & -1 & -1 \end{bmatrix}$$

TLaplace



Tratamiento de Imágenes

Operaciones Geométricas



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